

Wolf (Rex)

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Education

- Brigham Young University, Idaho – Computer Engineering major, Background in Physics, Emphasis in Applied Robotics

Skills & Abilities

HARDWARE AND EMBEDDED SYSTEMS

- Hardware and systems integration lead on team that designed and developed an autonomous terrain-mapping reconnaissance drone utilizing long distance telemetry and multisensory input including LiDAR, GPS and Barometrics
- Primary systems engineer and back up driver for high school robotics team that advanced to World Championships
- Experienced with micro-processors and IC's – Utilized Python, C++, C, and C#, to code and configure various devices including Arduino, Cortex, Raspberry pi, Node IC, MSP432 Launchpad, and a variety of STM32 project boards
- Knowledgeable in circuit design – Teaching assistant for PCB design and manufacturing for 4 semesters.

COMPUTER AND SOFTWARE

- Knowledgeable in system networking, computer and server hardware and integration, on both local and enterprise levels
- Assembled and rebuilt several private sever networks including NAS, Compute nodes, and multimedia encoding.

CAD AND 3D MODELING, DESIGN & VIDEO

- Knowledgeable in Autodesk Inventor, Auto-CAD, On Shape, Sketchup, Craftware, Cura, TrueNAS, OBS, and Adobe Suite
- Built and coded several 3D printers –Designed and printed multiple objects including a thermal transfer phone case
- Designed personal room lighting system using solar array and generator to be “off-the-grid”

OTHER SKILLS

- Hands-on learner, proficient in trade skills including electrical, drywall, plumbing and electrical/mechanical soldering
- Avid Camper, Hiker, Rock Climber, & Tree Feller. Scuba certified. Ham radio Technician.
- Winter sports enthusiast – Member of Beaverton High School Ski Team 2015-2017

LEADERSHIP

- Eagle Scout Dec 2017 – National Medal of Heroism 2019
- Build Team Leader and Systems Technician – Beaverton High School Robotics – 2016- 2018
- Served as Camp Director and Emcee for Boy Scouts of America Regional 11-year-old camp 2016-2018

Experience

TEACHER & LAB ASSISTANT | BYU IDAHO | 2022-2026

- Responsible for maintenance, repair and instruction of proper use of technical and electronic Lab equipment
- Student assistance for circuitry design and fabrication

SILICON RESEARCH GROUP | BYU IDAHO | 2025

- Designed, and fabricated mechatronic apparatus for controlled photolithography with accuracy on a micrometer scale
- Designed lensing apparatus to scale projected photo masks to the nanometer scale
- Designed and fabricated a control unit, to operate industrial goniometer for precise thin film measurements

PROSELYTING & SERVICE MISSIONARY | CHURCH OF JESUS CHRIST OF LATTER-DAY SAINTS | 2019-2020

- Member of Global Services operation providing help desk, network, database, & software support nationwide (2020)
- Served as proselyting missionary in Hawaii, Head Video Photographer and Social Media Director (2019)

TECHNICAL MANAGER | BEAVERTON HIGH SCHOOL | 2015-2018

- Responsible for operation and maintenance of all technical, electronic, and AV equipment

Interests

- Skiing, Rock Climbing, Scuba Diving, Long Distance Shooting, Bike Riding, Camping, Surfing, Archery, Robotics, Small Engine Mechanics, Solar Energy development, Computer/Server Networking, Ham Radio(Technician), Traveling